

Smart Factory AI Platform - Multi-Modal Industrial Intelligence System

This project is a web-based Multi-Modal Industrial Intelligence Platform that combines Machine Learning, Deep Learning, Computer Vision, Time-Series Forecasting, and Full-Stack Development. The system helps factories improve quality control, safety, maintenance, demand forecasting, and operational efficiency through a unified dashboard.

Module 1: Visual Quality Inspection

Problem: Detect defective products such as scratched metal sheets, cracked tiles, damaged bottles, and defective PCBs. AI Model: CNN + ResNet50. Datasets: MVTec AD, NEU Surface Defect. Output: Good/Defective classification with confidence score and highlighted defect regions using Grad-CAM.

Module 2: PPE Safety Detection

Problem: Detect workers not wearing helmets, safety vests, gloves, or other required equipment. AI Model: YOLOv8 object detection. Output: Real-time safety violation alerts and compliance monitoring.

Module 3: Predictive Maintenance

Problem: Predict machine failures before breakdown. Input Data: Temperature, vibration, pressure, RPM, and sensor readings. AI Model: LSTM and Autoencoder-based anomaly detection. Output: Failure probability, anomaly alerts, and estimated breakdown timeline.

Module 4: Demand Forecasting

Problem: Forecast future demand to reduce overproduction and inventory shortages. Input Data: Historical sales, inventory levels, seasonal trends. AI Model: Transformer or LSTM. Output: Future demand forecasts and production recommendations.

Module 5: Energy Optimization

Problem: Reduce energy consumption and operating costs. AI Model: Regression and Time-Series Forecasting. Output: Energy usage prediction, peak load estimation, and savings recommendations.

Multi-Modal Fusion Architecture

Image data is processed through CNNs, sensor data through LSTMs, and production logs through Transformers. Features are combined in a fusion layer to generate a Factory Health Score. Outputs include defect trends, machine health alerts, PPE violations, and operational recommendations.

Web Application Features

Dashboard: • Total Machines • Defect Rate • PPE Violations • Production Forecast • Energy Consumption Analytics: • Failure Trends • Defect Trends • Production Trends Reports: • Machine Health • Quality Metrics • Safety Metrics • Recommendations • PDF Report Generation

Technology Stack

Frontend: React, Tailwind CSS, Recharts Backend: FastAPI Machine Learning: Scikit-Learn, XGBoost Deep Learning: TensorFlow or PyTorch Computer Vision: OpenCV, YOLOv8, ResNet50 Database: PostgreSQL Deployment: Docker, AWS

Resume Value

This project demonstrates: • Machine Learning • Deep Learning • Computer Vision • Time-Series Forecasting • Full-Stack Development • Explainable AI • Industrial AI Applications • System Design and Deployment It is highly suitable for internships and AI/ML engineering portfolios.